

8

C

$$g = \frac{GM}{r^2}$$

$$g = \frac{G}{r^2}(\rho V) = \frac{G}{r^2}(\rho) \left(\frac{4}{3} \pi r^3 \right)$$

$$= \frac{4G\rho\pi}{3} r$$

$$g \propto r$$

Ratio of radius of X to Y $\rightarrow 3 : 1$

9

B

The speed of rotation of Earth is not related to gravitational potential energy.